

APURBA ROY

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Portfolio | GitHub | Leetcode | GeeksforGeeks

TECHNICAL SKILLS:

- **Languages:** Java, C, Python, Javascript, Typescript, HTML [Databases: MySQL, MongoDB, Firebase.]
- **Frameworks/Libraries:** React, RestAPIs, Tailwind CSS, Django, Node, Next, Express, Numpy, Pandas, JQuery, Bootstrap, Matplotlib, Tensorflow, Langchain, LLaMAIndex, HuggingFace, OpenAI SDK, Gemini API, Gradio/ Streamlit.
- **Developer Tools:** Docker, Kubernetes, Jupyter, MySQL, GCP, VSC, , Git, GitHub, Heroku, Figma.
- **Operating Systems:** Windows GUI, KaliLinux, GitBash, CMD, Powershell.

WORK EXPERIENCE:

- **Indian Cyber Security Solutions** Jan 2026 - Feb 2026
Cyber Security Internship
Conducted vulnerability assessment and penetration testing (VAPT) on web applications to identify security weaknesses. Used tools such as Nmap, Burp Suite, Wireshark, and Metasploit for reconnaissance, scanning, and basic exploit testing. Documented vulnerabilities and recommended mitigation strategies to improve system security.
- **IBM Skillbuild Virtual Internship** Dec 2025 - Jan 2026
AI & Cloud Traine
Gained hands-on experience with **IBM Cloud** services and tools. Developed practical skills through guided modules and projects focused on **cloud computing**, problem-solving, and real-world technology applications.
- **EXCELR** July 2025 - Aug 2025
Generative AI Internship
Worked on building and fine-tuning AI models for text and image generation. Getting practical experience in prompt engineering, LLMs, RAG, Langchain and integrating AI APIs into applications, able to develop real-world AI solutions
- **Techshaksham (AICTE) initiative by MICROSOFT & SAP** Jan 2025 - Feb 2025
ML Intern
A joint CSR initiative of Microsoft & SAP. It provides a unique opportunity for skills enhancement of students in the field of AI. I have been working on **"SMS Spam Detection using NLP"** under multiple Mentors for 4 weeks.
- **TATA Group** Dec 2024 - Jan 2025
Datavisualization Empowering Business (Virtual Internship)
Transformed raw datasets into actionable insights using **Power BI** and **Excel**, developing 5+ dashboards to track key performance indicators and inform strategic decisions for process efficiencies across workflows.

PROJECTS:

- **ElderGuard AI** Jan 2026 - Present
ElderGuard AI is a next-generation elder care monitoring platform designed to bridge the gap between aging independence and family oversight. By leveraging advanced AI and real-time data, we provide a safety net that respects privacy while offering immediate protection.
TECH STACK: React 18, Vite, TypeScript, Tailwind CSS, Microservices Architecture, Node.js, Python, AI/LLMs, Computer Vision, Firebase.
- **AI Interview Coach** Dec 2025 - Jan 2026
Built an AI-powered interview preparation tool that conducts mock interviews, generates role-specific questions, and provides intelligent feedback using IBM foundation models.
TECH STACK: Python, IBM watsonx, IBM Granite, LLMs, APIs, NLP.
- **MindSpace** Mar 2026 - Apr 2026
Developed an AI mental health companion featuring real-time mood detection, proactive crisis safety nets, and personalized relaxation strategies. Integrated Groq's low-latency LPU engine and LLaMA-3.3-70b to deliver highly empathetic, context-aware conversational support via a responsive UI.
TECH STACK: Python, Streamlit, Groq API (LLaMA-3.3-70b), SQLite, OAuth, Hugging Face Spaces (Deployment)

- Personalized Course Pathway using Agentic AI** **Aug 2025**
 A Generative AI-powered platform that creates and adapts personalized learning journeys for students — tailored to their goals, skills, and learning style. Built for students, educators, and EdTech innovators seeking scalable, intelligent course guidance using IBM Cloud.
TECH STACK: Python, RAG, LLM, GPT API, IBM Cloud, IBM Watson.
- SMS Spam Detection using NLP** **Jan 2025**
 SMS Spam Detection using NLP leverages natural language processing techniques to analyze and classify text messages as spam or legitimate. It enhances communication security by identifying patterns, keywords, and context in messages for effective and automated filtering.
TECH STACK: Python, NLP, Linear Regression, Jupyter.
- Smart Finance Tracker** **Nov 2024**
 Smart Finance Tracker is an intelligent system that automatically monitors, categorizes, and analyzes personal expenses and income patterns with smart budgeting suggestions and real-time insights .
TECH STACK: Javascript, Typescript, Solidity, Tensorflow.
- Face-Recognition-Attendance-system** **Aug 2024 - Sept 2024**
 A "Face-Recognition Attendance System" to automate attendance tracking through facial recognition. This system enhances efficiency and ensures secure, contactless attendance management for workplaces, school or events.
TECH STACK: Python, OpenCV, Tkinter, Dlib.

CERTIFICATIONS:

- MERN Essential Training** **Jan 2025**
 Linkedin Learning
- Career Essential AI Generative AI by Microsoft And Linkedin** **Mar 2025**
 Linkedin Learning
- Full stack JavaScript Development: MongoDB, Node, and React** **Feb 2025**
 Linkedin Learning
- React Hooks Crash Course** **Jan 2025**
 GreatStack
- MongoDB Developer's Toolkit** **July 2025**
 GeeksForGeeks
- Certified Machine Learning I Course by Columbia University** **Dec 2025**
 Columbia University

EDUCATION :

- Maulana Azad Kalam Abul University of Technology (SVIST)**
 2023 - 2027
 Pursuing Btech in CSE AI & ML
- St. Thomas Boy's School, Khiderpore**
 2011 - 2023
 10th C.G.P.A - 7.5 (2021) 12th C.G.P.A - 7.0 (2023 - Computer Science)

HACKATHONS & AWARDS:

- INNOVENTION 2026 (TECHFEST)** **April 2026**
 Secured **2nd place** at college techfest (**INNOVENTION**), leading the **ElderGuard AI** project; recognised by judges for innovation, real-world impact, and shift from reactive emergency systems to **predictive healthcare solutions**.
 - SMART INDIA HACKATHON** **Sept 2024**
 Made a project on "**Development of AI-ML based models for predicting prices of agri-horticultural commodities such as pulses and vegetable**". and came **1st RUNNER UP** in Internal Hackathon
 - NASA SPACE APPS CHALLENGE** **Oct 2024**
Winner of NSAC 2024 36 Hrs Hackathon from Durgapur Problem Statement "Symphony of the Stars: Harmonizing the James Webb Space Telescope in Music and Images". **GLOBAL NOMINEES** : Our team "**Super Coders**" Among **9,900** submitted projects only **940** projects has been selected .
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